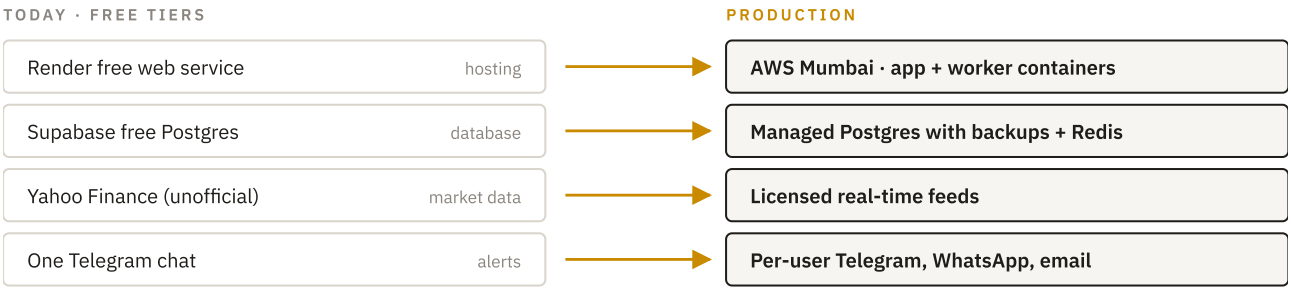
CODECAFE SIGNALS · VERSION 1 TO PRODUCTION

Production migration plan

Version 1 runs on free services. This document lists every service we use today, what each one moves to for a production build and why, the order of the moves, and what we need, from our side and from the client, to make each move.



01 Where version 1 stands

02 Why we have to move

03 Migration map: platform

04 Migration map: data and product

05 Migration plan in four phases

06 What we need to set up

07 Details needed from the client

08 Target architecture

01 · WHERE VERSION 1 STANDS

A working paper-trading signals desk on free services

Version 1 is live at `codecafe-signals.onrender.com`. Each logged-in user gets live market data, strategy signals with entry, stop-loss and target, a personal paper account (virtual money), trade history, a backtest, an AI news summary, the AI Screen Coach and risk protections. It is one Python service (Starlette + uvicorn) with background workers, deployed from a private GitHub repo. No real-money order is placed from the website.

SERVICE TODAY	WHAT IT DOES	FREE-TIER LIMIT
Render (free)	Hosts dashboard, API, background workers	Sleeps after 15 min idle; 512 MB RAM; slow cold start
Supabase (free)	Logins and paper accounts (one JSON row per user)	Pauses after ~7 days idle; 500 MB; no daily backups
Yahoo Finance	Gold, silver, oil, S&P, stocks, FX candles; search; news	Unofficial; CME futures ~10 min late; not licensed for commercial use
Bybit public API	Live crypto candles (polling)	Rate limits; spot only
Google Gemini · Groq	Screen Coach · news summary	Free request limits, no SLA
Google News RSS	Headlines and economic events	Unofficial feed
Telegram Bot API	Signal and SL/TP alerts	One chat for everyone
GitHub + Actions	Code and automated tests	Free minutes for private repos

02 · WHY WE HAVE TO MOVE

Six problems the free setup cannot solve

It goes to sleep

Render free stops after 15 idle minutes, so SL/TP checks and alerts stop with it.

Data can be lost

Supabase free pauses and keeps no daily backups. Accounts are JSON blobs, not tables.

Prices are late and unlicensed

Yahoo futures run ~10 min behind and cannot be resold in a paid product.

It cannot scale

One process, one in-memory lock, 512 MB. A second server would double-fill trades.

No product basics

No sign-up, password reset, payments, per-user alerts or monitoring.

Compliance

Paid signals in India need SEBI RA/IA status or an education-only model, plus DPDP duties.

03 · MIGRATION MAP: PLATFORM

From this, to that, and why

The recommended choice is shown first; the alternative below it is a like-for-like option if the client prefers it or already uses that provider.

AREA	TODAY	MOVE TO	WHY
App hosting	Render free web service	**AWS Mumbai (ap-south-1), ECS Fargate**, 2+ containers behind a load balancer Alt: Render paid Standard · DigitalOcean App Platform (Bangalore)	Always on, no sleep; low latency and data residency for Indian users; auto-scaling
Background jobs	Inside the web process	**Separate worker service** (ECS task) with a job queue (Redis) Alt: Render background worker	SL/TP settle, scans and alerts keep running during deploys and traffic spikes
Database	Supabase free, one JSON row per user	**AWS RDS PostgreSQL**, Multi-AZ, daily backups, point-in-time restore Alt: Supabase Pro	No pausing, no data loss; proper tables (users, accounts, trades, signals) with migrations (Alembic)
Cache and locks	In-memory lock, one process	**Redis** (AWS ElastiCache) Alt: Upstash Redis	Safe trade locks across servers; price cache; rate limits; job queue
Domain, SSL, edge	onrender.com subdomain	**Own domain on Cloudflare**: DNS, CDN, SSL, WAF, bot protection	Brand, trust, DDoS protection, reliable real client IP for rate limiting
Secrets	Render environment variables	**AWS Secrets Manager** Alt: Doppler	API keys rotated and audited, not copied by hand
Deploys	Push to main deploys straight to live	**GitHub Actions** builds a Docker image to ECR, deploys to **staging**, then to production on approval	Every change tested on a copy before clients see it; one-click rollback
Monitoring	None	**Sentry** (errors), **Better Stack** (uptime, status page, logs), CloudWatch metrics and alarms	We hear about failures before clients do
Backups	None	**Automated DB snapshots**, 30-day retention, monthly restore test	Recover from mistakes or outages
Code	GitHub private repo	**Stays on GitHub**; add branch protection and required reviews	No change needed

04 · MIGRATION MAP: DATA AND PRODUCT

Licensed data and the services a product needs

AREA	TODAY	MOVE TO	WHY
Gold, silver, oil, indices, FX	Yahoo CME futures, ~10 min late	**OANDA v20 API** (streaming spot prices) Alt: Twelve Data · Polygon.io · broker MT5 bridge	Real-time spot prices that match what traders see; commercial terms
Indian markets (NSE, BSE, MCX)	Yahoo, may lag	**Authorised vendor: TrueData or Global Datafeeds** Alt: broker APIs (Zerodha Kite Connect, Upstox, Dhan)	Exchange-licensed real-time data, required for a paid product
Crypto	Bybit REST polling	**Bybit and Binance WebSocket streams**	Tick-level prices, no polling, fewer rate limits
News and calendar	Google News RSS	**Finnhub** (news + economic calendar) Alt: Benzinga · Trading Economics	Licensed, structured feed with symbols and timestamps
AI (Screen Coach, news)	Gemini and Groq free tiers	**Paid Gemini API** with **Claude** as fallback; per-user usage limits	Higher limits, uptime, no data used for training
User accounts	Admin creates every login	**Self sign-up**, email verification, password reset, Google login, 2FA for admins	Clients onboard without us
Payments	None	**Razorpay** subscriptions (India) and **Stripe** (abroad)	Plans, renewals, invoices, GST
Alerts	One Telegram chat for everyone	**Per-user Telegram** link, **WhatsApp Business** (Meta Cloud API or Gupshup), **email** (AWS SES or Resend)	Each user gets their own alerts on their channel
Transactional email	None	**AWS SES** or **Resend** on the client's domain	Verification, password reset, receipts, reports
Mobile	Responsive web page	**PWA** (installable, push notifications), later **Capacitor** store apps	App-like experience without a second codebase
Analytics	None	**PostHog** (product usage) and an admin reports page	See who uses what, and where users drop off
Legal and compliance	None	**Terms, privacy policy, risk disclosure**, consent records (DPDP Act 2023); SEBI RA/IA route or education-only	Required before charging users

05 · MIGRATION PLAN IN FOUR PHASES

The order we move things in

Each phase leaves the platform working. Version 1 stays live on Render until phase 1 is verified on the new hosting, then DNS is switched and Render is retired.

PHASE 1

Stabilise

Always on, no data loss.

- Dockerise the app
- Set up AWS account, VPC, ECS, RDS, Redis
- Move Supabase data to RDS; JSON to proper tables
- Split web and worker
- Domain on Cloudflare
- Staging environment and deploy pipeline
- Sentry, uptime, logs, backups
- Switch DNS, retire Render and Supabase

NEEDS

AWS account with billing, domain access, Cloudflare account

PHASE 2

Real data

Real-time, licensed prices.

- OANDA streaming for metals, oil, indices, FX
- NSE/MCX vendor feed
- Crypto WebSocket streams
- Finnhub news and calendar
- Price cache in Redis
- Remove Yahoo from production

NEEDS

Data vendor contracts and API keys; markets list from client

PHASE 3

Product

Clients can sign up and pay.

- Sign-up, email verify, reset, Google login
- Razorpay and Stripe subscriptions
- Plans and feature limits
- Per-user Telegram, WhatsApp, email alerts
- PWA install and push
- Admin reports, analytics

NEEDS

Razorpay/Stripe KYC, WhatsApp Business verification, email domain, plan list

PHASE 4

Scale and comply

Ready for paying users at volume.

- Auto-scaling and load test
- Security review and penetration test
- Strategy R&D with walk-forward tests
- Legal pages and consent records
- SEBI route agreed
- Store apps if needed

NEEDS

Legal counsel, SEBI decision, launch date

CODE CHANGES THE MOVE REQUIRES

- `store.py`: JSON blob per user to tables, with Alembic migrations
- `paper.py`: in-memory trade lock to Redis or row lock
- `dashboard_server.py`: move `background_worker` into its own process
- `universal_data.py`: add OANDA, NSE vendor and WebSocket sources behind the same interface
- `notify.py`: per-user channels instead of one chat
- `auth.py`: sign-up, verification, reset, OAuth
- Rate limit to read the real client IP from Cloudflare only
- New: `Dockerfile`, infrastructure as code (Terraform), billing module

06 · WHAT WE NEED TO SET UP

Accounts, access and decisions

ITEM	WHAT EXACTLY	OWNER
Cloud account	AWS organisation with billing, IAM access for our team, Mumbai region	Client (we set up)
Domain and DNS	Product domain moved to Cloudflare; email DNS records	Client
Market data contracts	OANDA, NSE/MCX vendor, Finnhub: commercial licence covering paying users	Client signs, we integrate
Payment gateways	Razorpay and/or Stripe business accounts with completed KYC	Client
WhatsApp Business	Verified Meta business, approved message templates	Client
AI keys	Paid Gemini and Anthropic API accounts	Client or us
Monitoring	Sentry, Better Stack, PostHog accounts	Us
Data migration window	A short agreed slot to copy Supabase data and switch DNS	Both
Legal	Terms, privacy policy, risk disclosure; SEBI RA/IA advice	Client's counsel

07 · DETAILS NEEDED FROM THE CLIENT

Business and users

- Who the users are and where (India only or global)
- Expected users at launch and in one year
- Free, paid or subscription; features per plan
- Legal entity name for terms and invoices

Markets and trading

- Markets: commodities, NSE/BSE, F&O, US stocks, forex, crypto
- Timeframes: intraday, swing, positional
- Strategies or risk rules they want built in
- Paper trading only, or real orders later

Brokers and data

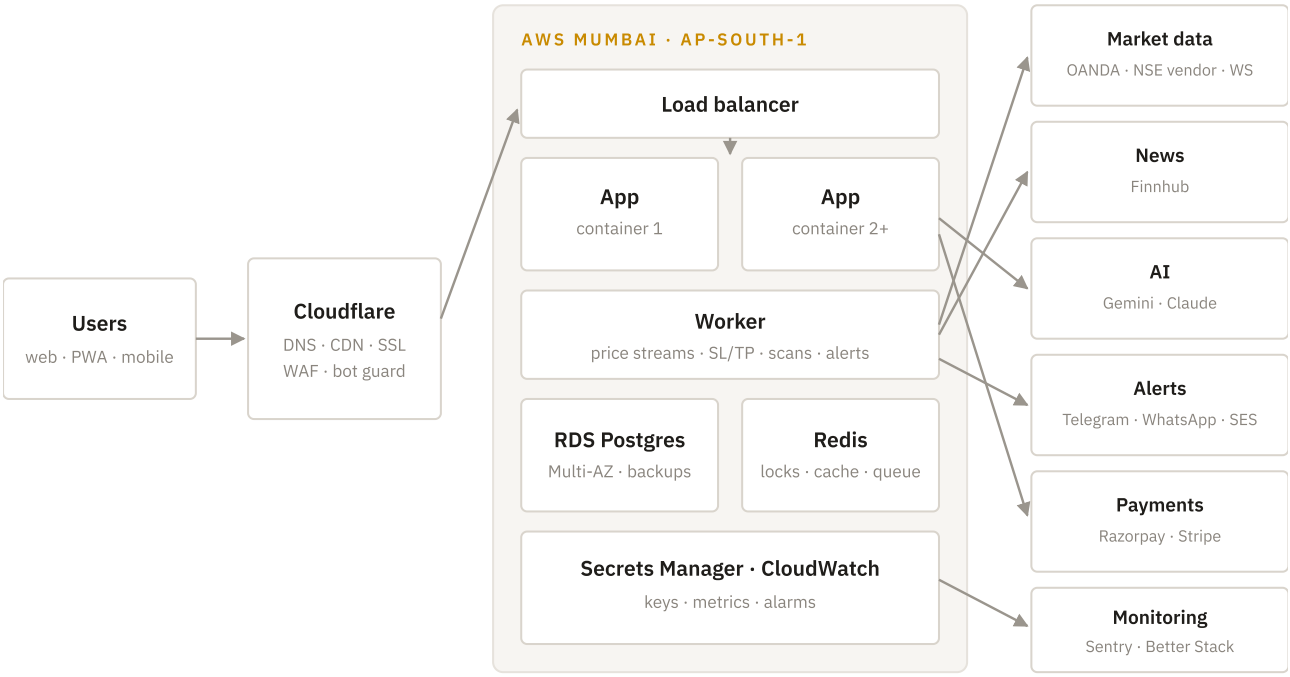
- Brokers used: Zerodha, Angel One, Upstox, Dhan, MT5, Bybit
- Existing data subscriptions or API keys
- Whether users connect their own broker accounts
- Acceptable data delay per market

Brand, compliance and access

- Product name, logo, colours, domain
- SEBI status or education-only plan
- Alert channels and admin reports needed
- Launch date and what "done" means

08 · TARGET ARCHITECTURE

What production looks like



GitHub Actions builds each release, deploys it to a staging copy, then to production on approval.

RISKS THE MIGRATION DOES NOT FIX

Strategy edge

The honest backtest shows no proven edge yet: gold 0% win, S&P 30%, Bitcoin 40%, small samples. Needs R&D before any performance claims.

Regulation

Selling signals in India needs SEBI RA/IA registration or an education-only product. Decide before phase 3.

Signal timing

Signals are on closed H1 bars, up to an hour after a move starts. Lower timeframes need the real-time feeds from phase 2.